Dynamical Systems

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Disclaimer

This document contains the lectures given by Dr. Chemllal.

To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

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Chapter 1

Discrete Dynamical Systems

1.1. Discrete Dynamical System

Definition 1.1.1 (Discrete Dynamical System): *A pair (X, F) is discrete dynamical system if X is a topological space and $F : X \to X$ is a continuous function.*

Definition 1.1.2 (Orbits): *Let (X, F) be a DDS and $x \in X$.*
- *Positive orbit of x is defined as $\mathcal{O}(x) = \{F^n(x) \mid n \in \mathbb{N}\}$.*
- *x is periodic if $\exists$ period $p \in \mathbb{N}^*, F^p(x) = x, \forall i < p, F^i(x) \neq x$.*
- *x is said to be eventually periodic if $\exists k \in \mathbb{N}, F^k(x)$ is periodic.*
- *The set of periodic points is denoted $\mathrm{Per}(F)$.*

Example:

- Rotations: let $X = [0, 1)$, the unit circle, and $R_\alpha : [0, 1) \to [0, 1), x \mapsto x + \alpha \bmod 1$

  Proposition 1.1.3: *There exists a no periodic point $\leftrightarrow \alpha$ is rational.*

- Doubling map: let $X = [0, 1)$ and $B : [0, 1) \to [0, 1), x \mapsto 2x \bmod 1$

Proposition 1.1.4: *$\mathrm{Per}(B)$ is dense.*

1.1.1. Dynamical Systems Of The Interval

We consider a dynamical system defined on $X = [a, b]$, we have the following theorem.

Theorem 1.1.1.5 (Sharkovskii): *Define the following order on n:*

$$2^k p \prec 2^{k'} p' \leftrightarrow k > k' \text{ or } (k = k' \text{ and } p < p')$$

which gives the following lexicographical order

$$\begin{array}{ccccccccccc}
3 & \succ & 5 & \succ & 7 & \succ & 9 & \succ & 11 & \succ & \cdots \\
2 \cdot 3 & \succ & 2 \cdot 5 & \succ & 2 \cdot 7 & \succ & 2 \cdot 9 & \succ & 2 \cdot 11 & \succ & \cdots \\
\vdots & & \vdots & & \vdots & & \vdots & & \vdots & & \\
2^n \cdot 3 & \succ & 2^n \cdot 5 & \succ & 2^n \cdot 7 & \succ & 2^n \cdot 9 & \succ & 2^n \cdot 11 & \succ & \cdots \\
\vdots & & \vdots & & \vdots & & \vdots & & \vdots & &
\end{array}$$

Then if f has a point of period n it has a point of all periods $m \succ n$.

1.1.2. Dynamical Systems On Symbol Spaces

Definition 1.1.2.6 (Bernoulli Shift): *Let A be a finite set of alphabets, we define the Bernoulli shift as the map $\sigma : A^{\mathbb{N}} \to A^{\mathbb{N}}, x \mapsto \sigma(x)$ with $\forall i \in \mathbb{N}, \sigma(x)_i = x_{i+1}$.*

An automaton is defined as a dynamical system $((\mathbb{Z}/p\mathbb{Z})^{\mathbb{N}}, O)$.

1.2. Lyapunov Stability

Definition 1.2.7 (Stable/Attractive): *Let (X, F) be a DDS and let $x_0 \in X$ is a fixed point.*
- *x is stable if $\forall \varepsilon > 0, \exists \delta > 0, \forall y \in B(x, \delta), \forall n \in \mathbb{N}, F^n(y) \in B(x, \varepsilon)$.*
- *x is attractive if $\exists \delta > 0, \forall y \in B(x, \delta), \lim_{n\to\infty} F^n(y) = x$.*
- *x is asymptotically stable if it is attractive and stable.*

1.2.1. Stability Of Interval Maps

Suppose that $F : [a, b] \to [a, b]$ a differentiable function, and x is a fixed point, then if $|F'(x)| < 1$ then x is asymptotically stable, if $|F'(x)| > 1$ then x is unstable.

1.2.2. Equicontinuity

Let (X, F) be a DDS, we say that x is an equicontinuity point if $\forall \varepsilon > 0, \exists \delta > 0, \forall y \in B(x, \delta), \forall n \in \mathbb{N}^*, F^n(y) \in B(F^n(x), \varepsilon)$. We say that (X, F) is equicontinuous if all its points are equicontinuity points.